



The TomMac Problem

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Tomass

Introduction

- ▶ TomMac processes 3 types of tomatoes into 4 distinct products
- ▶ Constraints in supply, production, demand and contracts
- ▶ Maximise annual profit

The Base Case Result

- ▶ The core financials:
 - ▶ Total annual income: £136,498.0
 - ▶ Total annual cost: £112,006.3
 - ▶ Maximised total annual profit: £24,491.7
- ▶ The total income breakdown:
 - ▶ Contract
 - ▶ Demand

Quarter	Contract(£)	Demand(£)	Total Income(£)
Jan-Mar	23,515	4,580	28,095
Apr-Jun	23,515	13,990	37,505
Jul-Sep	23,515	13,990	37,505
Oct-Dec	23,515	9,878	33,393

The Base Case Result

- ▶ The total cost breakdown:

- ▶ Tomato cost
- ▶ Production cost
- ▶ Inventory cost

Quarter	Tomatoes(£)	Making Products(£)	Storing(£)	Total Cost(£)
Jan-Mar	8,733.3	11,021.9	0	19,755.2
Apr-Jun	9,967.6	20,501.5	9	30,478.1
Jul-Sep	9,785.1	31,155.5	500	41,440.6
Oct-Dec	8,959.5	11,198.8	174.1	20,332.4

- ▶ Negative profit during Jul-Sep, indicating the increased purchase of stock to meet contract and demand needs for upcoming seasons.

The Base Case: Raw Material

- ▶ The quantity of tomatoes used for each quarter is:

Quarter	Beef(kg)	Plum(kg)	Moneymaker(kg)
Jan-Mar	1,500	4,183.3	4,000
Apr-Jun	2,643.6	5,544.8	9,000
Jul-Sep	4,207.8	8,403.9	13,000
Oct-Dec	1,500	4,409.5	4,000
Year	9,851.4	22,541.6	30,000

- ▶ The tomatoes supply for each quarter is:

Quarter	Beef(kg)	Plum(kg)	Moneymaker(kg)
Jan-Mar	1,500	5,000	4,000
Apr-Jun	4,000	10,000	9,000
Jul-Sep	6,000	15,000	13,000
Oct-Dec	1,500	5,000	4,000
Year	13,000	35,000	30,000

- ▶ From the two tables of tomato usage and supply we know:
 - ▶ All Moneymaker tomatoes are used and reaches the supply limit in each quarter.
 - ▶ All Beef tomatoes are used in 1st and 4th quarter.
 - ▶ Plum tomatoes are used in each quarter, but never reaches the supply limit.

The Base Case: Products

- The quantity of products being produced by each quarter is :

Quarter	Juice Production(kg)	Sauce Production(kg)	Paste Production(kg)	Canned Production(kg)
Jan-Mar	1,000	1,892	866.7	5,000
Apr-Jun	2,000	3,000	3,589.7	5,000
Jul-Sep	2,000	4,410.3	7,000	5,000
Oct-Dec	1,848.2	697.7	1,319.1	5,000
Year	6,848.2	10,000	12,775.4	20,000

- The quantity of products being sold by each quarter is :

Quarter	Juice sold(kg)	Sauce sold(kg)	Paste sold(kg)	Canned sold(kg)
Jan-Mar	1,000	2,000	2,500	5,000
Apr-Jun	2,000	3,000	3,500	5,000
Jul-Sep	2,000	3,000	3,500	5,000
Oct-Dec	1,848.2	2,000	3,275.4	5,000
Year	6,848.2	10,000	12,775.4	20,000

The Base Case: Products

- ▶ The quantity of products being held in stock by each quarter is :

Quarter	Juice in stock(kg)	Sauce in stock(kg)	Paste in stock(kg)	Canned in stock(kg)	Total in stock (kg)
Jan-Mar	0	0	0	0	0
Apr-Jun	0	0	89.7	0	89.7
Jul-Sep	0	1,410.3	3,589.7	0	5,000
Oct-Dec	0	108	1,633.3	0	1,741.3

- ▶ From the production, sale and stock information we have:
 - ▶ The warehouse is full at the end of 3rd quarter
 - ▶ Sales of canned tomatoes equal demand in all quarters
 - ▶ The juice and canned are the most popular products and are out of stock of all times
 - ▶ The paste is the most left product in stock

The Base Case: Profit Limits

- ▶ 1st quarter Jan-Mar:
 - ▶ Supply of Beef and **Moneymaker** is limiting the profit
 - ▶ Contract of Juice, Sauce and Paste is limiting the profit
 - ▶ Demand of **Canned** is limiting the profit
- ▶ 2nd Apr-Jun:
 - ▶ Supply of **Moneymaker** is limiting the profit
 - ▶ Demand of each product is limiting the profit

The Base Case: Profit Limits

- ▶ 3rd quarter Jul-Sep:
 - ▶ Supply of **Moneymaker** is limiting the profit
 - ▶ Demand of each product is limiting the profit
 - ▶ Production limit of paste is limiting the profit
 - ▶ Warehouse size is limiting the profit
- ▶ 4th quarter Oct-Dec:
 - ▶ Supply of Beef and **Moneymaker** are limiting the profit
 - ▶ Contract of Sauce is limiting the profit
 - ▶ Demand of **Canned** is limiting the profit

The Base Case: Profit Limits

- ▶ Summaries:
 - ▶ Supply of **Moneymaker** is limiting the total profit annually
 - ▶ Demand of **Canned** is limiting the total profit annually

Strategic decisions

- ▶ TomMac predicts increase of demand and overall product prices by 20%, and increase in production costs by 5%
- ▶ 3 strategic arrangements:
 - ▶ Rent a new warehouse of capacity 5,000kg at a cost of £3,000 per year
 - ▶ Sign a new contract which is specified in the table
 - ▶ Increase production limit by 20% at a cost of £400 per year
- ▶ New contract:

	Juice	Sauce	Paste	Canned
New contract (kg)	600	1,000	1,000	2,000
New contract (£/kg)	2.316	3.192	5.220	2.460

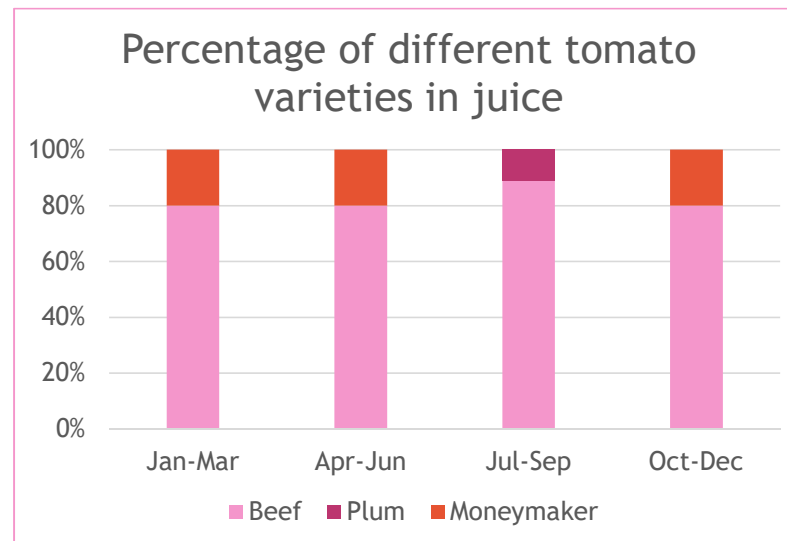
Strategic decisions

- ▶ Look for all possible combinations of arrangements: 8
- ▶ All but 2 strategies are possible for consideration
- ▶ When signing a new contract, renting a new warehouse is essential!
- ▶ The best strategy yields a profit of ~ £56,000
- ▶ Achieved by increasing warehouse and accepting the new contract

Strategy	Profit (£)	New Contract	New Warehouse	Incr. Production Lim.
1	52,445.7	No	No	No
2	52,745.7	No	No	Yes
3	53,978.0	No	Yes	No
4	54,313.0	No	Yes	Yes
5	Infeasible	Yes	No	No
6	Infeasible	Yes	No	Yes
7	56,041.7	Yes	Yes	No
8	55,884.5	Yes	Yes	Yes

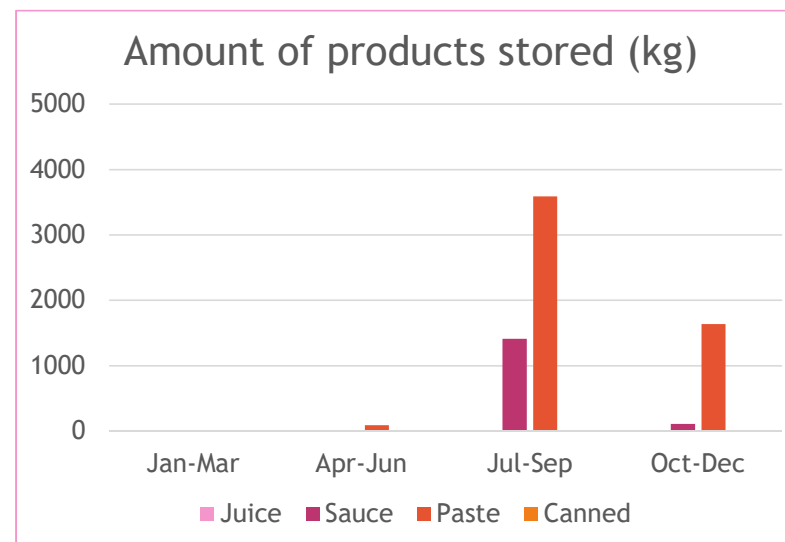
Potential Considerations

- ▶ Juice content varies quarterly
- ▶ Possible development of new product with different characteristics



Potential Considerations

- ▶ Products may change in quality during storage of several months
- ▶ Consider expiration dates, product quality



Potential considerations

- ▶ Emissions compliance
- ▶ More detailed product analysis
- ▶ Other expansions: new markets, new products, new tomato varieties

Conclusion

- ▶ Our model meets the current company policy in all the prescribed areas
- ▶ Most expensive product per kg: paste
- ▶ Most reliable product: canned tomatoes
- ▶ Most profitable change of current conditions:
 - ▶ demand for tomato paste
- ▶ Most consistent profit increase in current conditions:
 - ▶ Demand for canned tomatoes
 - ▶ Supply of moneymaker variety

Conclusion

- ▶ Most profitable strategic decision:
 - ▶ Do not sign new contract
 - ▶ Rent new warehouse
 - ▶ Increase production capacity
- ▶ Space for increasing future cooperation

Q&A Session